Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

- (●) VPC only
- ( ) VPC and more

Name tag - *optional*
Creates a tag with a key of 'Name' and a value that you specify.

test-vpc

IPv4 CIDR block Info

- (●) IPv4 CIDR manual input
- ( ) IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block Info

- (●) No IPv6 CIDR block
- ( ) IPAM-allocated IPv6 CIDR block
- ( ) Amazon-provided IPv6 CIDR block
- ( ) IPv6 CIDR owned by me

Tenancy Info

Default

VPC > Internet gateways > Attach to VPC (igw-05c65c97bbe2ad511)

Attach to VPC (igw-05c65c97bbe2ad511) Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

vpc-0319767e3917fe366

▸ **AWS Command Line Interface command**

Cancel | Attach internet gateway

Subnet 1 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

test-public-subnet-1a

The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Osaka) / apne3-az3 (ap-northeast-3a)

- No preference
- Asia Pacific (Osaka) / apne3-az3 (ap-northeast-3a) — ap-northeast-3-zg-1
  Type: availability-zone
- Asia Pacific (Osaka) / apne3-az1 (ap-northeast-3b) — ap-northeast-3-zg-1
  Type: availability-zone
- Asia Pacific (Osaka) / apne3-az2 (ap-northeast-3c) — ap-northeast-3-zg-1
  Type: availability-zone

▾ **Tags - *optional***

Key	Value - *optional*	
Name	test-public-subnet-1a	Remove

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

test-public-subnet-1b

The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Osaka) / apne3-az1 (ap-northeast-3b)

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.3.0/24 256 IPs

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - *optional*
Create a tag with a key of 'Name' and a value that you specify.

tes-public-RT

VPC
The VPC to use for this route table.

vpc-0319767e3917fe366 (test-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - *optional*	
Name	tes-public-RT	Remove

Add new tag

You can add 49 more tags.

VPC > Route tables > rtb-063ece631b906ec2f > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/2)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
test-public-subnet-1b	subnet-09969f38e173c5b08	10.0.3.0/24	–	Main (rtb-0ed37095e5a40bc86)
test-public-subnet-1a	subnet-0162f32aace615619	10.0.1.0/24	–	Main (rtb-0ed37095e5a40bc86)

Selected subnets

subnet-09969f38e173c5b08 / test-public-subnet-1b

subnet-0162f32aace615619 / test-public-subnet-1a

Cancel Save associations

VPC > Route tables > rtb-063ece631b906ec2f > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
10.0.0.0/16	local	Active	No	CreateRouteTable	
	local				
0.0.0.0/0	Internet Gateway	–	No	CreateRoute	Remove
	igw-05c65c97bbe2ad511				

Add route

Cancel Preview Save changes

Network settings Info

VPC - *required* | Info

vpc-0319767e3917fe366 (test-vpc)
10.0.0.0/16

Subnet | Info

subnet-0162f32aace615619 test-public-subnet-1a
VPC: vpc-0319767e3917fe366 Owner: 731451714946 Availability Zone: ap-northeast-3a (apne3-az3)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24)

Create new subnet

Auto-assign public IP | Info

Enable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

Security group name - *required*

launch-wizard-1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*

Description - *required* | Info

launch-wizard-1 created 2026-04-10T04:54:31.610Z

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0) Remove

User data - *optional* | Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash

yes | sudo apt update
yes | sudo apt install apache2

echo "<h1>Server Details</h1>
<p><strong>Hostname:</strong> $(hostname)</p>
<p><strong>IP Address:</strong> $(hostname -I | cut -d' ' -f1)</p>" \
| sudo tee /var/www/html/index.html

sudo systemctl restart apache2
```

☐ User data has already been base64 encoded

Not secure 15.168.13.247

Server Details

Hostname: ip-10-0-1-53

IP Address: 10.0.1.53

VPC | ap-northeast-3 | Instance details | EC2 | ap-north | 15.152.228.158

Not secure 15.152.228.158

Server Details

Hostname: ip-10-0-3-16

IP Address: 10.0.3.16

Create target group

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the ta

Settings - *immutable*

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.",

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

- (●) **Instances**
  Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
  Suitable for: ALB NLB GWLB
- ( ) **IP addresses**
  Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
  Suitable for: ALB NLB GWLB
- ( ) **Lambda function**
  Supports load balancing to a single Lambda function. ALB required as traffic source.
  Suitable for: ALB
- ( ) **Application Load Balancer**
  Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
  Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

test-ec2-alb-demo-tg

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 20/32

Protocol
Protocol for communication between the load balancer and targets.

HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

- (●) IPv4
  Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
- ( ) IPv6
  Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). Learn more

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0319767e3917fe366 (test-vpc)
10.0.0.0/16

Protocol version

- (●) HTTP1
  Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.
- ( ) HTTP2
  Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are

Available instances (2/2)

Filter instances

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-0539f34c9a8491dab	test-ec2-instance-2	Running	launch-wizard-2	ap-northeast-3b	10.0.3.16	subnet-09969f38e173c5b
i-0cbdc60dcfb72d094	test-ec2-instance-1	Running	launch-wizard-1	ap-northeast-3a	10.0.1.53	subnet-0162f32aace6156

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

2 selections are now pending below. Include more or register targets when ready.

Review targets

Targets (2)

Remove all pending

Filter targets

Show only pending

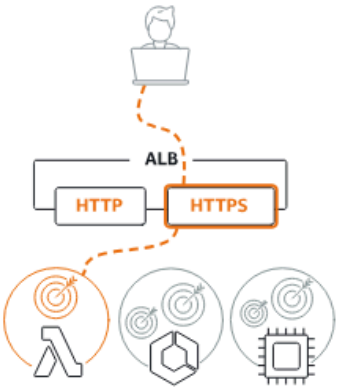
Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-0539f34c9a8491dab	test-ec2-instance-2	80	Running	launch-wizard-2	ap-northeast-3b	10.0.3.16	subnet-09969f38e173c5b08	April 10, 2026, 10:41 (UTC+05:30)
i-0cbdc60dcfb72d094	test-ec2-instance-1	80	Running	launch-wizard-1	ap-northeast-3a	10.0.1.53	subnet-0162f32aace615619	April 10, 2026, 10:26 (UTC+05:30)

Compare and select load balancer type

A complete feature-by-feature comparison along with detailed highlights is also available. Learn more

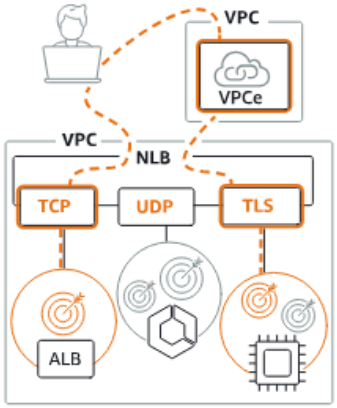
Load balancer types

Application Load Balancer Info



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and

Network Load Balancer Info



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per

Gateway Load Balancer Info

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

Create Application Load Balancer Info

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When
evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

▶ How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

test-lb-for-ec2-demo

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | Info

Scheme can't be changed after the load balancer is created.

- (●) Internet-facing
  - Serves internet-facing traffic.
  - Has public IP addresses.
  - DNS name resolves to public IPs.
  - Requires a public subnet.
- ( ) Internal
  - Serves internal traffic.
  - Has private IP addresses.
  - DNS name resolves to private IPs.
  - Compatible with the **IPv4** and **Dualstack** IP address types.

Load balancer IP address type | Info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

- (●) IPv4
  Includes only IPv4 addresses.
- ( ) Dualstack
  Includes IPv4 and IPv6 addresses.
- ( ) Dualstack without public IPv4

Network mapping Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.

vpc-0319767e3917fe366 (test-vpc)
10.0.0.0/16

Create VPC

IP pools | Info

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in the Amazon VPC IP Address Manager console.

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | Info

Select at least one Availability Zone and one subnet for each zone. We recommend selecting at least two Availability Zones. The load balancer routes traffic to targets in these Availability Zones only. Availability Zones that are not supported by the load b available for selection.

☑ **ap-northeast-3a (apne3-az3)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and configured to allow traffic.

subnet-0162f32aace615619 (test-public-subnet-1a)
10.0.1.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☑ **ap-northeast-3b (apne3-az1)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and configured to allow traffic.

subnet-09969f38e173c5b08 (test-public-subnet-1b)
10.0.3.0/24

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

Create security group

default (sg-0a4acacee628f9a36) ✕
Inbound rule (1) Outbound rule (1)

security-group-for-alb-demo (sg-09f4a37d50953d7c2) ✕
Inbound rule (1) Outbound rule (1)

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener **HTTP:80** Remove

Protocol: HTTP

Port: 80
1-65535

Default action | Info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

◉ Forward to target groups ○ Redirect to URL ○ Return fixed response

Forward to target group | Info

Choose a target group and specify routing weight or create target group.

Target group	Weight	Percent
test-ec2-alb-demo-tg Target type: Instance, IPv4 | Target stickiness: Off HTTP	1 0-999	100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness | Info

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

sg-for-alb

Name cannot be edited after creation.

Description Info

Allow access from internet

VPC Info

vpc-0319767e3917fe366 (test-vpc)

Inbound rules Info

Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
HTTP	TCP	80	Anyw...	0.0.0.0/0		Delete

0.0.0.0/0 ×

Introducing ALB target optimizer
Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. Learn more

test-lb-for-ec2-demo

Actions

Details

Load balancer type	Status	VPC	Load balancer IP address type
Application	Active	vpc-0319767e3917fe366	IPv4
Scheme	**Hosted zone**	**Availability Zones**	**Date created**
Internet-facing	Z5LXEXXYW11ES	subnet-09969f38e173c5b08 ap-northeast-3b (apne3-az1) subnet-0162f32aace615619 ap-northeast-3a (apne3-az3)	April 10, 2026, 11:00 (UTC+05:30)

Load balancer ARN
arn:aws:elasticloadbalancing:ap-northeast-3:731451714946:loadbalancer/app/test-lb-for-ec2-demo/c35f4c5646f34f5f

DNS name Info
test-lb-for-ec2-demo-895384586.ap-northeast-3.elb.amazonaws.com (A Record)

Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

---

Server Details

Hostname: ip-10-0-1-53

IP Address: 10.0.1.53

Server Details

Hostname: ip-10-0-3-16

IP Address: 10.0.3.16